Aymen Kabil

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Seeking a 6-month end-of-studies internship from April 2027 – distributed systems, cloud & backend

EDUCATION

Institut Polytechnique de Paris – Télécom SudParis 2026 – 2027
M2 Parallel and Distributed Systems (PDS) – track FODQ Palaiseau, France
- **Courses**: HPC (OpenMP, MPI, CUDA, Slurm), multicore programming (lock-free algorithms, NUMA), distributed algorithms

Télécom Paris – Institut Polytechnique de Paris 2025 – 2027
Diplôme d'Ingénieur (MSc in Engineering), double-degree programme with INPT Palaiseau, France
- **Tracks**: Distributed Software Systems (SLR); Maths, Theoretical CS & Operations Research (MITRO)

Institut National des Postes et Télécommunications (INPT) 2023 – 2025
Engineering cycle, Software Engineering – algorithms, cloud computing, software architecture Rabat, Morocco

Lycée Mohamed V – Classes Préparatoires (CPGE) 2021 – 2023
*Physics & Engineering Sciences – ranked **20th** / **912** at the national entrance exam (CNC)* Casablanca, Morocco

EXPERIENCE

CIEMS.ai June 2025 – Sept. 2025
Software Engineering Intern – AI-assisted vehicle damage assessment Rabat, Morocco
- Built the **Flask** REST backend of **CIEMS-AutoVision** for insurers: JWT auth, role-based access control, rate limiting
- Integrated **Gemini** and **GPT** vision models to pre-fill damage reports from vehicle photos
- Developed the **Next.js/React** frontend (claim-to-report workflow, PDF export); turned client interviews into specs

INPT – SEEDS Research Team (CEDoc 2TI) Feb. 2025 – May 2025
Research Intern – Software architecture optimisation Rabat, Morocco
- Designed **HARO**, a multi-objective optimiser proposing microservice decompositions of Java monoliths, ranked by ROI
- Modelled codebases as dependency graphs enriched with **CodeBERT** embeddings and Spring framework-level edge weights; implemented the full Python pipeline
- Benchmarked 20+ methods (Louvain, Leiden, GA, simulated annealing, tabu search, NSGA-II) on 5 open-source Spring systems

Ministry of Health June 2024 – Sept. 2024
Software Engineering Intern – national youth health platform (ages 10–25) Rabat, Morocco
- Developed the "Information & Education for Health" module (Spring Boot, PostgreSQL, Kafka, Docker); ran requirement meetings with stakeholders in Scrum sprints

PROJECTS

Quorum – replicated key-value store on Raft, written from scratch – Go, Docker, Prometheus GitHub 2026
- Deterministic Raft core and fsync'd write-ahead log; safety checked over 1,000 randomized fault seeds (loss, partitions, crash-restarts), linearizability verified with Porcupine
- Durable writes from 842 to ∼3,450/s (p50 18.8 → 4.7 ms) via pipelining, group commit and connection pooling, re-validated by the fault suites

Vigil – fleet telemetry & failure detection – Go, Python, MQTT, Kafka, Prometheus GitHub 2026
- Streaming pipeline for a fleet of simulated machines: MQTT ingestion with per-device Last Will, Go gateway to Kafka (keyed per machine, backpressure), Prometheus/Grafana instrumentation, CI
- Phi-accrual failure detector adapting to each machine's heartbeat rate, distinguishing dead connections from stalled applications

Fault-Tolerant Key-Value Store – Java, Akka actors, Python (team of 4) GitHub 2026
- Quorum-based multi-writer/multi-reader atomic register (ABD-style) over N Akka actors, tolerating $f < N/2$ crashes; linearizability of histories checked, latency measured over 9 scenarios

AWARDS, LEADERSHIP & ACTIVITIES

Highest Kaggle score (30 teams) – HAKS Hackathon (Airbus × IBM), corrosion-risk prediction 2026

2nd place, grand final – ASC Systems Engineering Challenge (IP Paris), system architecture design 2025 – 2026

Vice-President, INPT Chess Club – organised internal tournaments and weekly training sessions 2024 – 2025

Member, English Debating Club, Télécom Paris 2025 – present

SKILLS & LANGUAGES

Programming: Java, Python, Go, C, C++, TypeScript/JavaScript, SQL
Systems & Backend: Spring Boot, Akka, Flask, Node.js, REST, Apache Kafka, Redis, PostgreSQL, MySQL, OpenMP, MPI
DevOps & Cloud: Docker, Git, GitHub Actions, Prometheus, Grafana, Linux
AI & Research: LLM integration (Gemini, GPT), NLP, graph algorithms, multi-objective optimisation
Languages: English – full professional (TOEIC 955/990); French – C1 (DALF C1); Arabic – native
Interests: Chess (1900+ rapid on chess.com), competitive programming, mathematics